Research on Driving Factors of COVID-19 Protective Behaviours in Australia Based on Deep Learning

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Abstract

During the COVID-19 pandemic, personal protective behaviour, especially wearing masks, played a crucial role in controlling of the infection. Although public health policies such as mask mandates significantly changed people's compliance, understanding the behavioural driving factors behind protective behaviours are still important. These driving factors are complex and dynamically influenced by macro pandemic factors such as vaccination rates and confirmed cases. This study investigates the determinants of COVID-19 protective behaviours in Australia through a two-phase machine learning framework. In the first phase, this study reproduced Ryan et al.'s (2025) research, using random forest and XGBoost to classify and predict the protective behaviours before and after the mask mandates. The evaluation results showed that prior to the implementation of the mandates, people's protective behaviour was mainly driven by existing protective habits and frequency of non-household contact. After the mandates, the passage of time (week number) and an individual's willingness to isolate became the main predictive factors. To address the structural limitations of traditional tree models in capturing high-order and nonlinear variable interactions, the upcoming second phase proposes the construction of a Multi-layer Perceptron (MLP) architecture and the integration of state-level weekly vaccination rates and confirmed cases. In addition, to overcome the 'black box' characteristics of neural networks, this study will introduce SHAP (SHapley Additive exPlanations) values to quantify feature contributions.

Acknowledgements and Declarations

I would like to clarify the use of external tools and references in this project. After independently drafting the written content of this report, I used tools such as Grammarly and NetEase strictly to check for spelling and grammatical errors.

I maintained some same variable naming as Ryan et al. (2025) to get clear communication and direct comparison. Regarding the visualisations, I have used AI tools to extract the specific colour codes from Ryan et al.'s (2025) research for the feature importance figures to directly compare. Besides these, all final code was written by myself and does not contain any AI-generated scripts.

1 Introduction

During the COVID-19 pandemic, individual protective behaviours played a crucial role in controlling the spread of the virus (Guo et al., 2024). Among all protective behaviours, mask wearing was initially recognized as one of the most effective measures (Eikenberry et al., 2020). To cope with future pandemics, understanding the drivers of protective behaviours, particularly mask wearing, is essential for the design of public health policies.

There were many studies investigating why people wear masks from various perspectives (Ryan et al., 2025). Among these, mandated policy is one of the most important factors, which may lead to a sharp increase in the use of masks (Haischer et al., 2020). Recently, Ryan et al. (2025) investigated the role of face mask mandates in influencing the drivers of protective health behaviours in Australia. However, some key factors have not been considered. For example, the introduction of vaccination or COVID-19 case numbers may alter individuals' risk perception and behavioural responses (De Nicolò et al., 2022). There is a need to understand how different external factors drive behaviour change to inform future public health policy-making.

In order to identify these complex relationships, appropriate modelling approaches are needed. Most existing research relies on traditional statistical or tree-based models. Although tree-based models often perform strongly on tabular data, they may be limited in fully capturing complex interactions, particularly when multiple heterogeneous factors are involved (Rana et al., 2023).

To address these limitations, this study is structured into two phases. The first phase involved reproducing the analysis of Ryan et al. (2025) to understand the research framework and potential issues of this field. The reproduction phase has been completed. In the second step, this study integrated vaccination data and COVID-19 case numbers, and will apply deep learning methods to better capture complex relationships and identify the key drivers of protective behaviours, especially mask wearing.

For the whole study, the primary research questions are:

1. How do vaccination rates and COVID-19 case numbers influence the uptake of protective behaviours in Australia?

2. How does the performance of deep learning models compare to traditional statistical or tree-based models in predicting protective behaviours?

2 Background and Related Work

Previous studies have shown that individual protective behaviours are influenced not only by demographic and psychological factors (such as age and risk perception), but are also significantly influenced by policies (Wismans et al., 2022). In particular, mask mandates have been shown to change key drivers of behaviours. Ryan et al. (2025) found that protective behaviours are not fixed, but vary depending on the surrounding policy environment.

Furthermore, vaccination rates and COVID-19 case numbers are also important factors affecting protective behaviours, and may influence behaviours in more complex ways. For example, Andersson et al. (2021) found a phenomenon called 'risk compensation' after vaccination, which decreases perceived risk and influences protective behaviours.

These findings suggest that protective behaviours are influenced by multiple interacting external factors, implying that the relationships may be complex, and therefore require more advanced modelling approaches. Many existing studies use tree-based methods such as random forest and XGBoost, which often perform well on structured data (Uddin & Lu, 2024). The success of these non-linear models suggests that simple linear models may not be sufficient to fully capture behavioural patterns. However, tree-based models may still be limited in capturing more complex interactions between different factors, especially when multiple external factors are considered together. In contrast, Multilayer Perceptrons (MLPs) have shown potential in capturing complex nonlinear relationships and variable interactions (Bagheri et al., 2023). With appropriate regularization and parameter tuning, MLPs can achieve performance comparable to or even better than tree-based models in complex structured data tasks (McElfresh et al., 2023).

In the field of modelling protective behaviour, understanding the driving factors behind the predicted results is as important as the accuracy of the prediction itself. Model interpretation directly provides a basis for policy formulation. Although advanced machine learning architectures such as XGBoost and MLPs show excellent predictive performance when processing complex structured data, their 'black box' characteristics pose significant challenges in public health research (Fernández et al., 2018).

To address this lack of interpretability, studies are increasingly using SHAP (SHapley Additive exPlanations) values to quantify the contribution of various factors to the model's prediction results. SHAP values were proposed by Lundberg and Lee (2017), and their mathematical foundation is based on cooperative game theory (Shapley, 1953). This method treats each feature (such as mask mandates, vaccination rates, demographic factors, etc.) as one of the 'players' and the model's predicted results as the final 'payout'. By calculating the marginal contribution of each feature across all possible feature arrangements and combinations, SHAP provides a theoretically unified framework for explaining complex models.

Therefore, this study extends the analysis of Ryan et al. (2025) by introducing vaccination data and COVID-19 case numbers, applying an MLP model, and finally utilizing SHAP values for model interpretation.

3 Methodology

This section details the experimental design and implementation methods of the study. The research adopts a two-phase framework that begins with a reproduction of the study by Ryan et al. (2025) and continues with an extended experimental phase.

In terms of data source composition, the reproduction phase used two core datasets, while the extension phase expanded the scope to four datasets. Because the preprocessing pipelines for the shared core datasets are the same, to ensure a logical and concise structure, this section will first provide an overview of the background and cleaning methods of all four datasets, and then detail the modelling strategies corresponding to different stages in sections.

3.1 Data Sources

To explore the research questions, this study used four datasets. This section introduces the sources and contents of the data. For detailed processing procedures, please refer to the section 3.2.

3.1.1 YouGov Behaviour Tracker Dataset

The Imperial College London YouGov Behaviour Tracker dataset (Jones et al., 2020) contains the self-reported survey responses of individuals from 29 countries regarding protective health behaviours against COVID-19 between 2020 and 2022. The survey included demographic and psychological factors (such as gender and age), protective behaviours (such as mask wearing and social distancing), comorbidities, and more features. As this study focuses on the situation in Australia, only the Australian subset of this dataset is used.

3.1.2 Oxford COVID-19 Government Response Tracker Dataset

The Oxford COVID-19 Government Response Tracker dataset (Hale et al., 2021) records the state level response policies of governments around the world aimed at preventing the spread of COVID-19 from March 2020 to January 2023. The recorded policies include containment and closure, economic, healthcare, vaccination, and other general policies. This dataset is used for the mandate period split.

3.1.3 Pandemic Case Data

Reporting of COVID-19 cases varied across each state and territory in Australia. Hence, this study uses a consolidated dataset (covid19data.com.au,

2023) for COVID-19 case numbers, which compiles and verifies COVID-19 case data from Australian federal, state, and territory health departments. The data records daily confirmed cases at state and territory level granularity, covering the period from January 2020 to August 2023. It is only used in the extension phase.

3.1.4 Vaccination Data

Since the Australian government has not released detailed vaccination data for each state on a weekly basis, this study relies on a vaccine dataset compiled by a third-party team (covid19data.com.au, 2023) to match the timing and locations of the YouGov survey. This dataset collects weekly vaccination totals for each state and territory from official health websites, covering January to August 2022. It is only used in the extension phase.

3.2 Data Cleaning, Variable Construction, and Data Preprocessing

This study preprocessed the datasets based on the methodology of Ryan et al. (2025) to meet the input requirements of the models. The specific feature conversion and cleaning steps are divided into eight core steps.

3.2.1 Government Mandates

During the COVID-19 pandemic, Australia experienced a variety of enforced orders. Due to the different response measures among local governments, the experience of COVID-19 in each state varied significantly. Therefore, the start time and enforcement of mask mandates varied greatly between different states. In order to evaluate the actual effectiveness of government mandates, this study aligned the data based on the daily mask wearing policy index (0 - 'No policy' to 4 - 'Required at all times') from the OxCGRT dataset.

Following Ryan et al.'s (2025) method, and considering the varying levels and effects of policy implementation, this study calculated the 14-day rolling average of each state's index. When the rolling average of a certain state reached 3 for the first time, it was considered that the jurisdiction had actually entered a stable period of mandate intervention. Then, the completion time of the individual survey was compared to their regions, and a variable `during_mandate` was generated for each observation.

3.2.2 Removal of Features with High Missingness

For the YouGov dataset, due to multiple item changes during the tracking period of the survey in Australia, some questions were only asked within an extremely short window of time. To prevent the loss of massive amounts of valid samples during the cleaning phase due to high missingness in a few features, this study counted the number of missing values for each feature, and same as Ryan et al.'s (2025) study, set 10,781 as the threshold. Columns with missing values greater than this threshold were considered not in the survey for a long period and were dropped from the dataset.

3.2.3 Repairing Data Gaps Caused by Survey Changes

During the data exploration process, it was found that YouGov modified the informed consent terms of the survey between February 10 and October 18, 2021. This led to a significant structural data gap in health related questions during this period. Deleting these records would mean losing eight months of important data. By keeping them as 'N/A', this study can save other useful responses, and improve models' performance. Therefore, this study located this time window using a date mask and filled in these missing values as independent 'N/A' categories, thus preserving valid samples.

3.2.4 Processing of Predictive Variables

The core prediction targets are self protection behaviour and mask wearing. The original survey records the frequency of respondents performing a certain behaviour in the past 7 days. This study first constructed a dictionary and measured responses on a five-point scale (from 1 - 'not at all' to 5 - 'always'). Then, features were divided into three logical branches:

- **Mask wearing behaviours:** There are four specific scenarios related to masks ('Worn a face mask outside your home', 'Worn a face mask inside a grocery store/supermarket', 'Worn a face mask inside a clothing/footwear shop', and 'Worn a face mask on public transportation'). 'Mask wearing behaviour' was defined as the median of all scenarios and named `face_mask_scale`. After that, 4 (i.e. 'frequently') was used as the compliance threshold to binarise this feature into a classification label of Yes/No, ensuring a consistent binary outcome definition with the methodology of Ryan et al. (2025).

- **Protective behaviours:** All self protection scenario scores with the prefix `i12_` were extracted, including other protective measures from the survey, such as social distancing, hand hygiene, and coughing covering. Their overall median was calculated as `protective_behaviour_scale`. The same threshold of 4 or above was used to binarise them into `protective_behaviour_bi`.

- **Protective features other than masks:** Ryan et al. (2025) showed that a person's protective habits could strongly predict whether they would wear a mask. To avoid confounding effects from respondents' other hygiene habits when predicting mask behaviour, this study specifically excluded the four mask scenarios mentioned above, calculating the median of all remaining protective behaviours as a continuous feature.

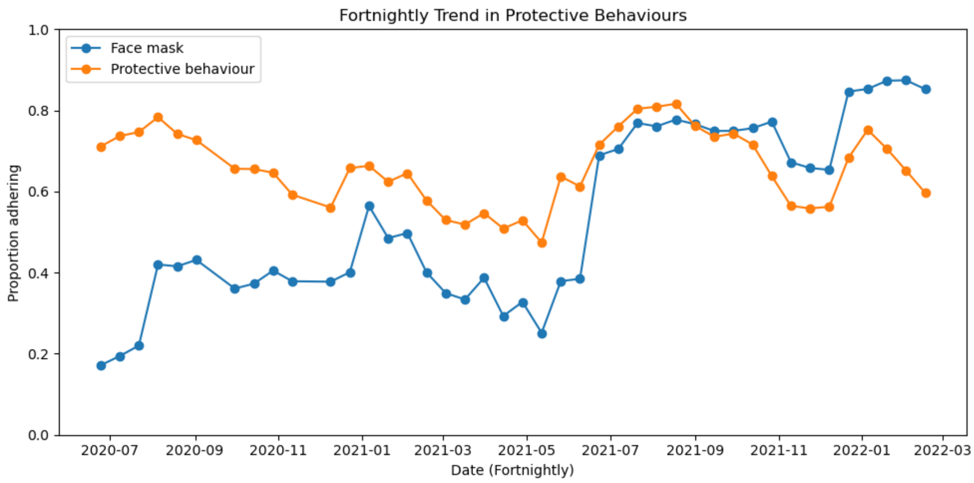

Figure 1: Public compliance between July 2020 and March 2022. The chart tracks two metrics: the face mask 'Yes' rate (blue) and the protective behaviours other than masks 'Yes' rate (orange).

Figure 1 shows the changes in people's protective behaviour from 2020 to 2022 after cleaning and preprocessing. Initially, the protective behaviours other than masks (orange line) was higher than the proportion of mask wearing (blue line). However, in mid-2021, both classes showed a significant increase. In the later stages, the proportion of people wearing masks exceeded other protective behaviours, reaching a high point of nearly 90%. Both variables exhibit similar temporal trends.

- **Other protective predictors:** Additionally, this study added three other self-protective features. The first is the average number of non-household contacts (`i2_health`), which was retained as a continuous numerical variable. The remaining two are subjective attitude features: willingness to isolate if unwell (`i9_health`) and willingness to isolate if directed to do so by a healthcare professional (`i11_health`). Since these subjective attitudes were recorded as categorical responses, they were transformed into dummy variables prior to modelling.

3.2.5 Dimensionality Reduction for Sparse Comorbidity Features

In the raw data, comorbidities were scattered among 13 binary options (from `d1_health_1` to `d1_health_99`), making the data extremely sparse. To improve model efficiency, this study removed all specific disease types and made them into a single variable called `d1_comorbidities`. Respondents who selected 'None of these' were recorded as 'No', 'Prefer not to say' was recorded as 'Prefer_not_to_say', records within the consent gap window were recorded as 'N/A', and any other confirmed diseases were grouped as 'Yes'.

3.2.6 Processing of Demographic Variables

The demographic variables used in this analysis include age, gender (male/female), state (one of the eight states or territories in Australia), employment status, and household size. When dealing with household size, the raw data exhibited a clear long-tail distribution, mixed with text responses. To reduce the effect of extreme values on models, this study assigned a value of 8 to all samples answering '8 people or more'. Invalid samples answering 'I don't know' or 'I don't want to disclose' were directly deleted at the end of the cleaning process.

3.2.7 Processing of Perception, Trust, and Mental Health Variables

In addition to demographic and behavioural data, this study included subjective variables such as perception of illness threat, trust in government, mental health, and wellbeing.

For perception of illness threat (perceived susceptibility and severity), the original responses were measured on a 7-point scale mixed with text (e.g., 7 - 'Agree' and 1 - 'Disagree'). This study cleaned the text characters and converted them into continuous variables. Wellbeing (Cantril Ladder), measured on a 10-point scale, was directly retained as a continuous numerical feature. Finally, for unordered categorical features such as trust in government, PHQ-4 mental health features, and comorbidities, dummy coding was applied with `drop_first = True` to remove the baseline group and prevent multicollinearity.

3.2.8 Vaccination and COVID-19 Case Numbers

To support the extension phase of this study and explore the potential driving effect of the macro public health environment on personal protection behaviour, two external datasets will be introduced in extension work: weekly COVID-19 case numbers and vaccination data in each state/territory. To ensure the rigor of reproducing Ryan et al.'s (2025) baseline study, this macro-level data was exclusively used during the extension phase and did not participate in the first phase's model training. In the expansion work, this study will use the `endtime` of the respondents' survey completion and their `state` value to merge the individual survey records with the corresponding state's vaccination rates and daily new cases per 100,000 population. This standardisation can solve the demographic differences across states.

3.3 Data Splitting and Scenario Construction

After completing data cleaning and feature engineering, the data was split into training and testing sets in an 80:20 ratio, stratified by the value of `during_mandate`. To evaluate the effectiveness of policies across multiple dimensions and ensure the reliability of model training, this study constructed a specific data splitting and cross-validation mechanism.

3.3.1 Construction of Predictive Scenarios

In both steps of this study, there were two main predictive objectives: 'mask wearing' and 'general protective behaviour'. To better analyse the impact of mandates on the main driving factors, this study used the `during_mandate` variable to establish six independent prediction scenarios. Initially, a global baseline model covering the entire time range was established for both targets (Model 1 and Model 2) , using the `during_mandate` variable as an independent predictor. Then, using the value of `during_mandate`, the datasets were strictly divided into before mandates (1a and 2a) and during mandates (1b and 2b). Training models on these subsets allows for the direct observation of how deterministic features evolve before and after mandates.

3.3.2 Class Imbalance and Resampling

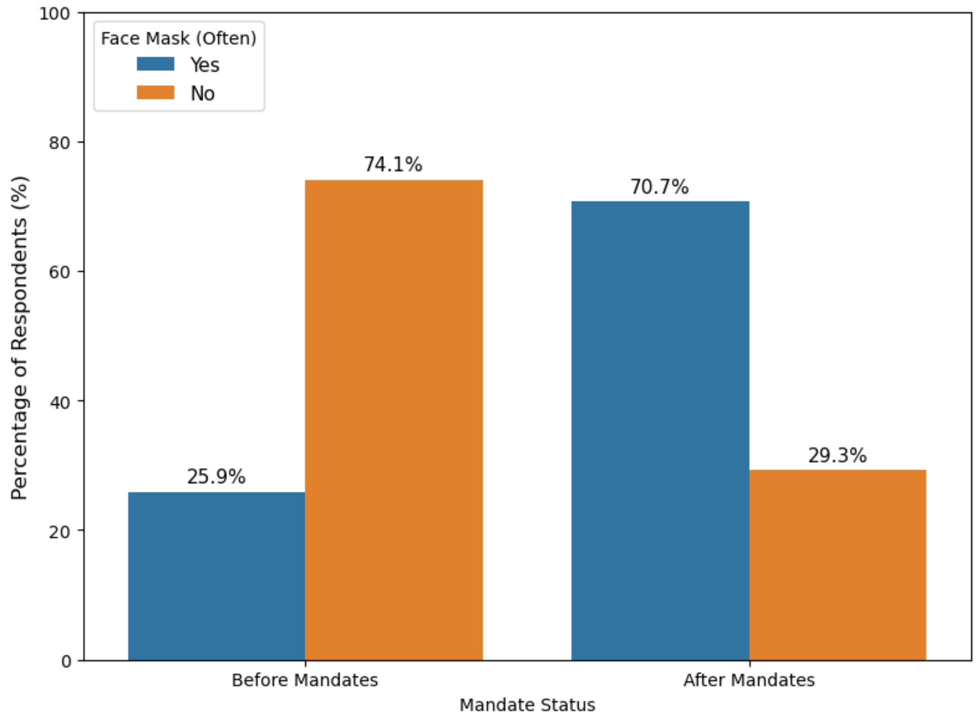

Figure 2: The proportion of respondents frequently wearing face masks before and after the implementation of mask mandates. The blue bars represent individuals who frequently wore face masks (Yes), where as the orange bars represent those who did not (No). The chart illustrates the extreme class imbalance present in the dataset, where the majority class reverses following the introduction of mandate policies.

As shown in Figure 2, before the mandates, the proportion of people who did not wear masks was as high as 74.1%, while those who frequently wore masks accounted for only 25.9%, demonstrating an extreme class imbalance. After mandates, this proportion reversed (70.7% wearing vs. 29.3% not wearing). If such imbalanced data is directly used for training, machine learning models may inherently predict the majority class, resulting in low capture ability for the minority class. To overcome this, this study introduced the random over sampler technique to deal with class imbalance.

3.3.3 Cross Validation

In terms of model validation strategy, a single validation set is highly susceptible to the randomness of the initial data split. To prevent overfitting during model training, this study implemented a 5-fold stratified shuffle split. This cross-validation mechanism ensures a robust and unbiased evaluation of model performance.

3.4 Modelling

In the reproduction phase, this study constructed four types of machine learning models using the Scikit-Learn and XGBoost libraries in a Python 3 environment. PyTorch will be used to establish the MLP model in the extension work. As mentioned, because of the extreme class imbalance, all classifiers were encapsulated within an Imblearn resampling pipeline during training. Similar to Ryan et al.'s (2025) study, the final feature importance is determined jointly by the two best-performing models.

3.4.1 Logistic Regression

Logistic regression is a generalised linear model widely applied in behavioural analysis, that outputs the probability of a target class. Serving as the baseline model, an L2 regularization penalty was introduced to prevent overfitting in the high-dimensional feature space. Specifically, the inverse of regularization strength (parameter C) was searched across a grid of [0.1, 1.0, 10.0].

3.4.2 Classification Trees

Classification trees are supervised learning methods that classify data by constructing a series of binary splits based on entropy or Gini impurity (Breiman et al., 1984). This model offers high efficiency and interpretability. Joint tuning was performed on the maximum tree depth `max_depth` ([5, 10, 15, None]), the minimum samples required at a leaf node `min_samples_leaf` ([1, 5, 10]), and the minimum samples required to split an internal node `min_samples_split` ([2, 5, 10]). Feature importance is calculated based on the total reduction of impurity brought by each feature across all split nodes.

3.4.3 Random Forests

Random Forest is an ensemble method that employs bagging (bootstrap aggregating) (Breiman, 1996). Random Forest generates multiple independent decision trees by resampling the training data with replacement while evaluating a random subset of features at each split node. These features ensure that the results are not completely dominated by strong individual classifiers. Hyperparameter tuning focused on the total number of trees `n_estimators` ([100, 250, 500]), maximum tree depth `max_depth` ([10, 20, None]), and the maximum number of features `max_features` ([‘sqrt’, ‘log2’]) to control feature subspace randomness.

3.4.4 XGBoost

XGBoost (Extreme Gradient Boosting) is a highly scalable ensemble boosting algorithm. Unlike the parallel structure in random forests, it fits new trees by leveraging the residual information from previous models to continuously optimise subsequent predictions, a process that defines the boosting mechanism (Chen & Guestrin, 2016). As a boosting algorithm sensitive to parameter configurations, tuned hyperparameter included `learning_rate` ([0.05, 0.1, 0.2]), column subsample ratio `colsample_bytree` ([0.6, 0.8, 1.0]), number of estimators `n_estimators` ([100, 250, 500]), and maximum tree depth `max_depth` ([4, 6, 8]).

3.4.5 MLP Model

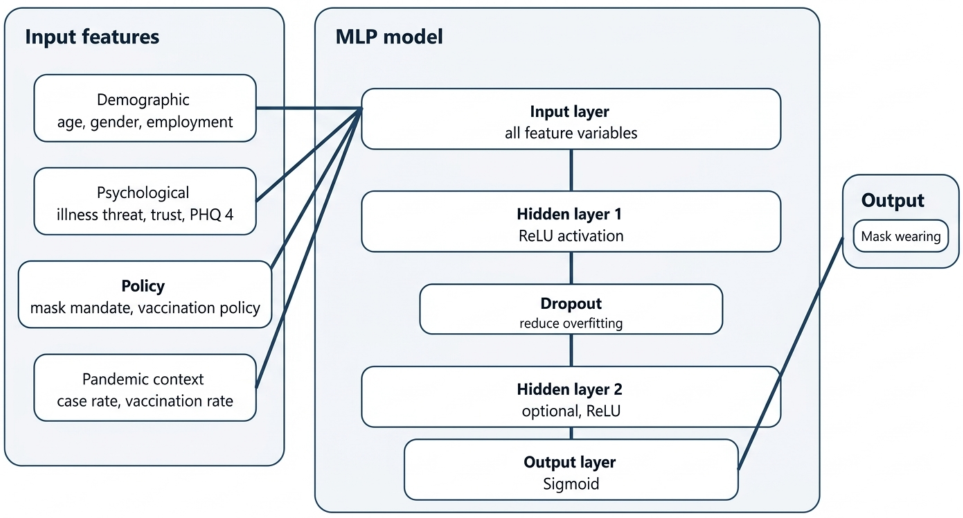

Figure 3: The structure of the MLP model in this study

In order to capture more complex non-linear relationships between protective behaviour and features, the extension phase introduces a Multilayer Perceptron (MLP) model(Sarker, 2021). As showen in Figure 3, all feature variables are fed into the input layer, which passes through one or two hidden layers utilizing nonlinear activation functions (e.g., ReLU). Dropout layers are included to mitigate overfitting. The output layer uses a sigmoid function to predict the probability of mask wearing as a binary outcome. The model is trained using the Adam optimiser(Kingma & Ba, 2014). The dataset contains 40,420 observations, which is sufficient to train an MLP, though likely inadequate for an extremely deep network. Therefore, the model initializes with a relatively simple structure, and the structure may be changed based on subsequent performance tuning.

3.5 Model Selection and Evaluation

3.5.1 Model Selection and Grid Search Cross-Validation

Similar to the baseline literature (Ryan et al., 2025), the reproduction section compares the performance of logistic regression, classification trees, random forests, and XGBoost on the initial dataset. Rather than utilizing Optuna, this study adopts a more intuitive `GridSearchCV` during the hyperparameter tuning stage. This method allowed for the definition of a comprehensive hyperparameter grid prior to tuning. To find the optimal parameter combination, a 5-fold Stratified Shuffle Split was applied to the training set, with the grid search explicitly configured to maximise the Area Under the Receiver Operating Characteristic Curve (ROC AUC) as the final refit metric. Since this workflow demonstrated strong performance during reproduction, a similar workflow will be created for the extension phase, incorporating the MLP model and integrating the cross-validation steps directly into the Imblearn pipeline.

3.5.2 Evaluation Metrics

During the model training and testing phases, multiple performance indicators were recorded. In addition to AUC, which evaluates the model's overall ability during tuning, this study also compared precision, recall, accuracy, and F1 score. Accuracy measures the proportion of correctly predicted behaviours, recall evaluates the model's ability to capture true positive samples, and the F1 score provides a harmonic mean of precision and recall, reflecting comprehensive performance in class-imbalanced environments (Fernández et al., 2018).

3.5.3 Model Interpretation

In Ryan et al.'s study, models were run 10,000 times to eliminate importance fluctuations caused by random up-sampling. Because the reproduction work focuses primarily on validating the methodological process rather than perfectly reproducing the original metrics, and due to computational limitations, this study did not perform 10,000 resampling iterations. Instead, feature importance was directly extracted from the best model (`best_estimator_`) identified by `GridSearchCV` (where the model was refitted on the entire training set using the best parameters). Since random forest and XGBoost performed best, their built-in `feature_importances_` attributes were used.

In the future extension work, the introduction of the MLP model needs an external interpretation mechanism due to its 'black box' architecture. Therefore, this study will introduce SHAP values to quantify feature contributions. This method will be applied to interpret the deep learning framework and the top-performing ensemble model for comparative analysis. Additionally, since upsampling will be used in the training pipeline, a stability analysis of the SHAP values will be conducted.

4 Results

This section demonstrates and explains the performance of the models, the results of model selection, and the comparison of key predictors before and after the mandate periods.

4.1 Model Performance and Selection

Table 1: Performance metrics of predictive models across different scenarios, with the standard error (SE) in parentheses. Results are extracted from the best model identified by GridSearchCV, and was refitted on the entire training set. Bold values show the optimal models based on largest AUC

Algorithm	AUC	Precision	Recall	Accuracy	F1 score
Face mask behaviour before mandates					
Logistic regression	0.816 (0.004)	0.480 (0.004)	0.753 (0.007)	0.723 (0.003)	0.587 (0.005)
Decision tree	0.811 (0.003)	0.476 (0.006)	0.773 (0.011)	0.718 (0.006)	0.589 (0.003)
Random forest	0.853 (0.004)	0.662 (0.009)	0.570 (0.008)	0.812 (0.004)	0.613 (0.008)
XGBoost	**0.855 (0.005)**	0.606 (0.008)	0.664 (0.008)	0.800 (0.005)	0.634 (0.008)
Face mask behaviour during mandates					
Logistic regression	0.836 (0.001)	0.880 (0.001)	0.777 (0.002)	0.768 (0.001)	0.826 (0.001)
Decision tree	0.848 (0.002)	0.880 (0.001)	0.828 (0.005)	0.799 (0.003)	0.853 (0.003)
Random forest	0.890 (0.002)	0.879 (0.002)	0.914 (0.002)	0.851 (0.002)	0.896 (0.001)
XGBoost	**0.897 (0.002)**	0.901 (0.002)	0.873 (0.003)	0.843 (0.003)	0.887 (0.002)
General protective behaviour before mandates					
Logistic regression	0.731 (0.003)	0.686 (0.002)	0.692 (0.002)	0.669 (0.002)	0.689 (0.002)
Decision tree	0.729 (0.004)	0.691 (0.005)	0.698 (0.012)	0.674 (0.003)	0.694 (0.005)
Random forest	**0.785 (0.004)**	0.721 (0.003)	0.743 (0.003)	0.711 (0.003)	0.732 (0.003)
XGBoost	0.782 (0.003)	0.727 (0.002)	0.726 (0.003)	0.710 (0.002)	0.727 (0.002)
General protective behaviour during mandates					
Logistic regression	0.787 (0.002)	0.864 (0.002)	0.728 (0.004)	0.721 (0.004)	0.791 (0.003)
Decision tree	0.778 (0.002)	0.859 (0.002)	0.743 (0.008)	0.726 (0.004)	0.797 (0.004)
Random forest	0.843 (0.001)	0.846 (0.001)	0.883 (0.003)	0.800 (0.002)	0.864 (0.002)
XGBoost	**0.850 (0.002)**	0.876 (0.002)	0.829 (0.002)	0.792 (0.001)	0.852 (0.001)

Following the hyperparameter tuning and 5-fold cross validation procedures detailed in Section 3.5, the predictive models were evaluated.Based on the evaluation results presented in Table 1, several macro trends consistently appear across all scenarios. The random forest and XGBoost significantly outperformed the classification tree and logistic regression in overall predictive ability. Compared to general protective behaviours, the results of all mask wearing models were better.

Table 2: Final evaluation metrics on the independent test set across different scenarios. Bold values show the optimal models based on largest AUC

Algorithm	AUC	Precision	Recall	Accuracy	F1 score
Face mask behaviour before mandates					
Random forest	**0.863**	0.652	0.613	0.819	0.632
XGBoost	**0.863**	0.603	0.711	0.808	0.653
Face mask behaviour during mandates					
Random forest	0.882	0.874	0.918	0.847	0.895
XGBoost	**0.885**	0.894	0.882	0.841	0.888
General protective behaviour before mandates					
Random forest	**0.793**	0.724	0.751	0.717	0.737
XGBoost	0.792	0.737	0.745	0.724	0.741
General protective behaviour during mandates					
Random forest	0.841	0.849	0.887	0.802	0.868
XGBoost	**0.849**	0.877	0.837	0.796	0.857

Public health mandates clearly influenced model efficacy. Before the mandates began, the evaluation metrics for all algorithms were generally lower, primarily because very few individuals were doing protective behaviours at that time. When predicting mask wearing before the mandate, both ensemble models maintained a higher AUC of approximately 0.85. Random Forest achieved better precision (0.662), while XGBoost exhibited superior recall (0.664).

During the mandate periods, performance improved significantly across all metrics. XGBoost achieved the highest AUC scores across all scenarios (e.g., reaching 0.897 for face mask wearing during mandates). Additionally, random forest excelled at generating precise predictions, achieving high F1 scores for both face mask behaviour (0.896) and general protective behaviour (0.864) during mandates.

Subsequently, the final predictive performance of random forest and XGBoost was validated in the final testing set (Table 2). The results closely mirrored the macro trends observed from the refitting on the training set. Consistent with the training phase, predictive performance was lower before mandates and improved significantly after their implementation. Furthermore, the models maintained their distinct performance advantages in testing data. XGBoost still achieved the highest AUC scores in face mask usage during mandates (0.885), while random forest still achieved the highest F1 scores in all during mandate scenarios. These results suggested that the models did not overfit the training data and have great generalization capabilities.

Overall, the models achieved robust predictive performance, with random forest and XGBoost alternating as the top performers depending on the specific metric and scenario.

4.2 Comparison of Key Predictors Before and After Mandate Periods

Due to the high impact of the 'state' variable, it was excluded from the visual plots to better illustrate the shifting dynamics of the remaining external and behavioural variables before and after the mandates. For comparative purposes, the specific color codes for feature categories was extracted from Ryan et al.'s (2025) research with the help of an AI tool(Nano Banana).

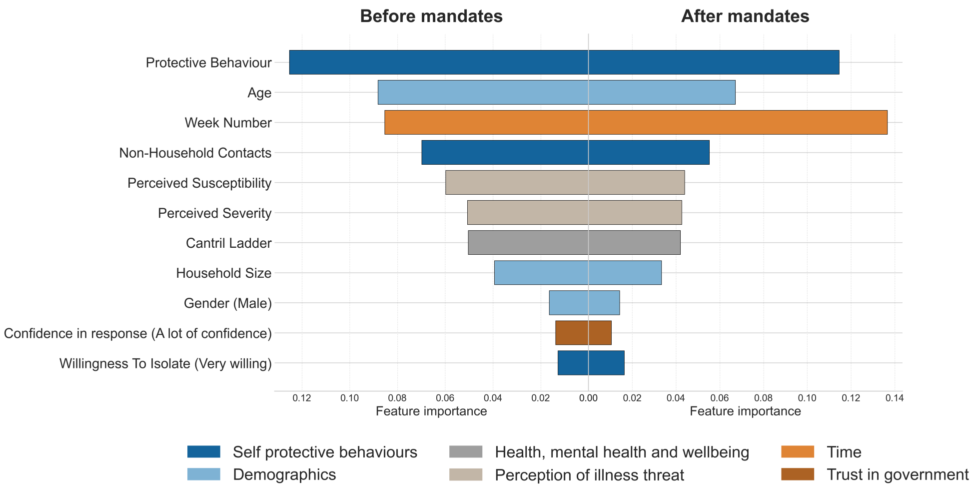

Figure 4: The top influential features when predicting face mask wearing, other than state, using a random forest model. Feature importance was directly extracted from the optimal model identified through GridSearchCV, which was refitted on the entire training set. The left hand panel displays the feature importance before face mask mandates started and the right hand panel shows the feature importance after face mask mandates start. The bars are coloured based on the category of the feature as defined in the legend.

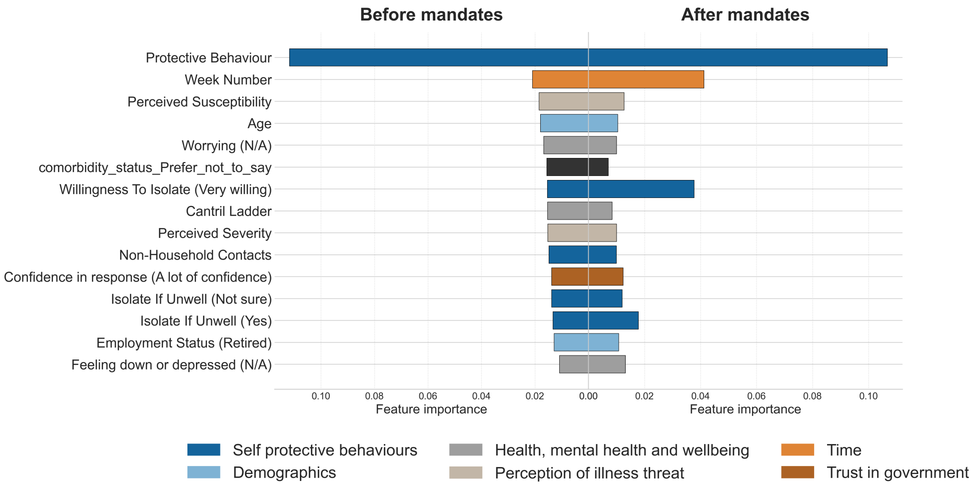

Figure 5: The top influential features when predicting face mask wearing, other than state, using an XGBoost model. Feature importance was directly extracted from the optimal model identified through GridSearchCV, which was refitted on the entire training set. The left hand panel displays the feature importance before face mask mandates started and the right hand panel shows the feature importance after face mask mandates start.

4.2.1 Face Mask Wearing

As shown in Figure 4, before the implementation of official mandates, protective behaviour and age were the top two predictors in the random forest. Interestingly, the XGBoost (Figure5) heavily highlighted protective behaviour, with its relative importance far exceeding all other features. Week number also played significant roles in both models.

The introduction of mandates did not significantly change the core composition of the driving factors for mask-wearing, though it did cause minor changes in rankings. After mandates, week number became an important feature, ranking first in the random forest and second in XGBoost. A notable shift in the XGBoost model was the rise in the importance of willingness to isolate, while protective behaviour remained crucial across both models. Demographics such as age and household size continued to contribute, but their relative influence was decreased compared to the pre-mandate period.

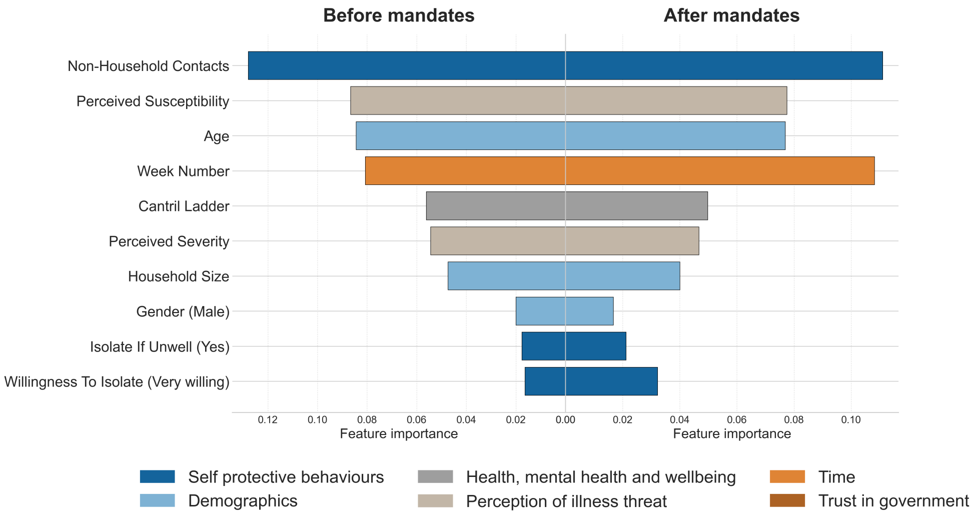

Figure 6: The top influential features when predicting general protective behaviours, other than state, using a random forest model. Feature importance was directly extracted from the optimal model identified through GridSearchCV, which was refitted on the entire training set. The left hand panel displays the feature importance before mandates started and the right hand panel shows the feature importance after mandates start. The bars are coloured based on the category of the feature as defined in the legend.

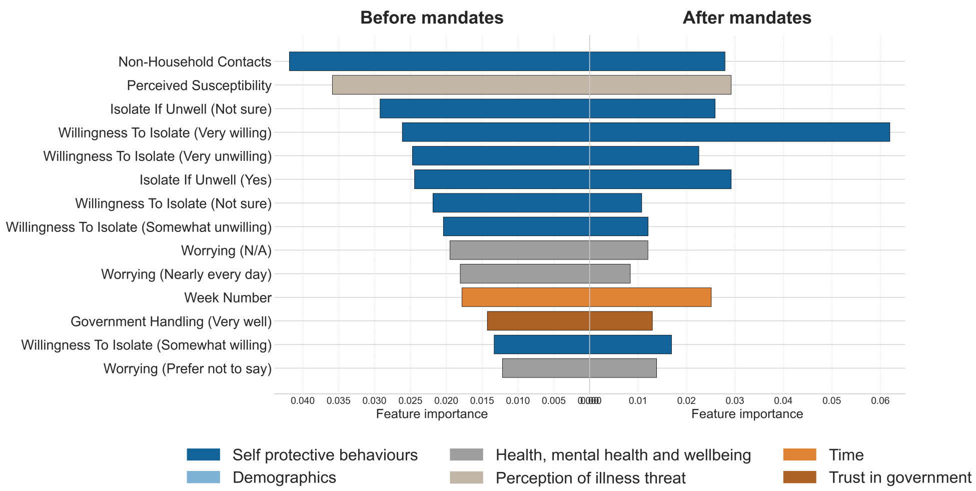

Figure 7: The top influential features when predicting general protective behaviours, other than state, using an XGBoost model. Feature importance was directly extracted from the optimal model identified through GridSearchCV, which was refitted on the entire training set. The left hand panel displays the feature importance before mandates started and the right hand panel shows the feature importance after mandates start. The bars are coloured based on the category of the feature as defined in the legend.

4.2.2 General Protective Behaviours

For general protective behaviour before the mandates, Figure 6 and Figure 7 indicate that non-household contacts were the most important feature across both models. In the random forest (Figure 6), the subsequent top features were perceived susceptibility, age and week number. In the XGBoost (Figure 7), perceived susceptibility ranked second, and also driven by several isolation-related features such as Isolate if unwell(not sure) and varying degrees of willingness to isolate.

After mandates, the ranking of some features shifted. In the random forest model (Figure 6), week number surged to become one of the most important features, ranking just below non-household contacts. Age and perceived susceptibility still remained within the top four. In the XGBoost (Figure 7), willingness to isolate (very willing) jumped to the highest rank, far exceeding the importance scores of all other variables.

5 Discussion

This study successfully reproduced the core predictive framework of Ryan et al. (2025) and validated the significant impact of public health mandates on driving factors to protective protective behaviours. Mask wearing increased significantly after mandates began. This matches other research showing that strict policies increase public compliance (Haischer et al., 2020). Based on the evaluation metrics, the current models' performance in predicting mask-wearing and protective behaviours before mandates was lower than that of Ryan et al.'s (2025) study. But after mandates, predictions slightly outperformed. These performance discrepancies are likely driven by the inherent randomness within the models and the variance introduced by the upsampling. Consistent with Ryan et al.'s (2025) study, the ensemble methods (random forest and XGBoost) significantly outperformed the logistic regression and classification trees. While the classification tree failed to surpass the logistic regression, the strong performance of tree-based ensembles on tabular data matches the findings of broader machine learning research (Uddin & Lu, 2024).

In terms of feature importance, the results captured a behavioural shift that broadly same as the findings of Ryan et al.(2025). The dominant role of protective behaviours other than mask in predicting mask wearing, alongside the increased importance of week number during mandate periods, was consistently confirmed. However, a significant deviation occurred regarding the exact ranking and distribution of these feature importance. These differences may be caused by this study extracted feature importance directly from a single optimal model rather than calculating a median across 10,000 random fits, the results may be more susceptible to localized variances.

During the reproduction phase, random forest and XGBoost demonstrated robust predictive capabilities on tabular data. To achieve better estimates in the upcoming expansion phase, this project will introduce macro-level pandemic data, specifically state-level COVID-19 case numbers and vaccination rates. Due to the fact that some pandemic factors can alter an individual's risk perception (e.g., the 'risk compensation' effect following vaccination (De Nicolò et al., 2022)), traditional tree-based architectures may have structural limitations when dealing with such highly complex, high-order variable interactions (Bagheri et al., 2023).

For future work, I will use a deep learning model called Multilayer Perceptron (MLP) to better capture complex relationships in the data set. To solve the lack of neural networks' interpretability, this study will use SHAP values to explain how the model works. While the first phase only provided a general ranking of important factors, SHAP can show exactly how much each specific detail influences a single prediction. Additionally, since upsampling will be used in the training pipeline, a stability analysis of the SHAP values will be conducted. Together, the MLP model and SHAP will give us both accurate results and a clear explanation of how the driving factors of behaviour shifted across the population.

6 Appendices

Code (ljj_pipeline_corrected.ipynb) and important data sets used for this study can be accessed via the following GitHub repository:

[https://github.com/payy906/JIAJUN-LI-code-for-Data-Science-Research-Project-Part-A-.git]

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